

- Gemma Team. 2024a. [Gemma 2: Improving open language models at a practical size](#). *Preprint*, arXiv:2408.00118.
- Gemma Team. 2024b. [Gemma: Open models based on gemini research and technology](#). *Preprint*, arXiv:2403.08295.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. 2023. [Llama: Open and efficient foundation language models](#). *Preprint*, arXiv:2302.13971.
- Eduard Tulchinskii, Kristian Kuznetsov, Laida Kushnareva, Daniil Cherniavskii, Sergey Nikolenko, Evgeny Burnaev, Serguei Barannikov, and Irina Piontkovskaya. 2023. Intrinsic dimension estimation for robust detection of ai-generated texts. *Advances in Neural Information Processing Systems*, 36:39257–39276.
- Adaku Uchendu, Zeyu Ma, Thai Le, Rui Zhang, and Dongwon Lee. 2021. [TURINGBENCH: A benchmark environment for turing test in the age of neural text generation](#). In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 2001–2017.
- Ben Wang and Aran Komatsuzaki. 2021. GPT-J-6B: A 6 Billion Parameter Autoregressive Language Model. <https://github.com/kingoflolz/mesh-transformer-jax>.
- Yuxia Wang, Artem Shelmanov, Jonibek Mansurov, Akim Tsvigun, Vladislav Mikhailov, Rui Xing, Zhuohan Xie, Jiahui Geng, Giovanni Puccetti, Ekaterina Artemova, Jinyan Su, Minh Ngoc Ta, Mervat Abassy, Kareem Ashraf Elozeiri, Saad El Dine Ahmed El Eter, Maiya Goloburda, Tarek Mahmoud, Raj Vardhan Tomar, Nurkhan Laiyk, Osama Mohammed Afzal, Ryuto Koike, Masahiro Kaneko, Alham Fikri Aji, Nizar Habash, Iryna Gurevych, and Preslav Nakov. 2025. [GenAI content detection task 1: English and multilingual machine-generated text detection: AI vs. human](#). In *Proceedings of the 1st Workshop on GenAI Content Detection (GenAIDetect)*, pages 244–261, Abu Dhabi, UAE. International Conference on Computational Linguistics.
- Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. 2021. [mt5: A massively multilingual pre-trained text-to-text transformer](#). *Preprint*, arXiv:2010.11934.
- Xianjun Yang, Liangming Pan, Xuandong Zhao, Haifeng Chen, Linda Ruth Petzold, William Yang Wang, and Wei Cheng. 2024. [A survey on detection of LLMs-generated content](#). In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 9786–9805, Miami, Florida, USA. Association for Computational Linguistics.
- Xiao Yu, Yuang Qi, Kejiang Chen, Guoqiang Chen, Xi Yang, Pengyuan Zhu, Weiming Zhang, and Nenghai Yu. 2023. Gpt paternity test: Gpt generated text detection with gpt genetic inheritance. *CoRR*.
- Rowan Zellers, Ari Holtzman, Hannah Rashkin, Yonatan Bisk, Ali Farhadi, Franziska Roesner, and Yejin Choi. 2019. [Defending against neural fake news](#). *Advances in Neural Information Processing Systems (NeurIPS)*, 32.
- Aohan Zeng, Xiao Liu, Zhengxiao Du, Zihan Wang, Hanyu Lai, Ming Ding, Zhuoyi Yang, Yifan Xu, Wendi Zheng, Xiao Xia, Weng Lam Tam, Zixuan Ma, Yufei Xue, Jidong Zhai, Wenguang Chen, Peng Zhang, Yuxiao Dong, and Jie Tang. 2023. [Glm-130b: An open bilingual pre-trained model](#). *Preprint*, arXiv:2210.02414.
- Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, Todor Mihaylov, Myle Ott, Sam Shleifer, Kurt Shuster, Daniel Simig, Punit Singh Koura, Anjali Sridhar, Tianlu Wang, and Luke Zettlemoyer. 2022. [Opt: Open pre-trained transformer language models](#). *Preprint*, arXiv:2205.01068.

A Related Work

Machine-Generated Text Detection. Detection systems for distinguishing human and AI-generated text follow two main approaches: Training-Based and Zero-Shot methods. Training-based approaches fine-tune Transformer models on labeled datasets for strong in-domain performance (Chen et al., 2023; Li et al., 2023; Yu et al., 2023). In contrast, zero-shot methods analyze statistical patterns without supervised fine-tuning, like token likelihoods, probability curvature or intrinsic dimension (Gehrmann et al., 2019; Mitchell et al., 2023; Tulchinskii et al., 2023).

However, the challenge of making AI-generated text more interpretable for humans has only been addressed by a limited number of approaches, either through manual analysis (Guo et al., 2023) or only partially investigating the dependencies (Kuznetsov et al., 2024).

Sparse Autoencoders and Interpretability. LLM interpretability is especially challenging due to polysemanticity, where a single neuron encodes multiple unrelated concepts (Elhage et al., 2022a, 2023). Sparse Autoencoders (SAEs) were proposed to help isolating more interpretable latent dimensions (Sharkey et al., 2023). Unlike standard autoencoders, SAEs introduce a penalty (e.g. L_1 regularization) to ensure that only a small subset of neurons is active per input, resulting in highly interpretable features (Cunningham et al., 2023).

Recent approaches use large language models or heuristics to automate hypothesis generation and refinement (Bricken et al., 2023; Cunningham et al., 2023; Gao et al., 2023). For example, (Bricken et al., 2023) employ GPT-4 to label sparse dimensions based on top-activating tokens, while (Cunningham et al., 2023) use heuristic methods like measuring overlap with linguistic categories to infer dimension meanings. In our work we employ both manual and automatic interpretation to ensure unbiasedness of our approach.

Datasets and Benchmarks. AI text detection includes many datasets, starting with GPT-2 Output (Solaiman et al., 2019) and Grover (Zellers et al., 2019), as well as TuringBench (Uchendu et al., 2021), which unifies 19 models for cross-evaluation. Additionally, domain-specific corpora and “in-the-wild” tests, such as (Chakraborty et al., 2023), become useful for enhancing model robustness.

B COLING dataset: additional details

The COLING dataset contains generations of the models from the following families: a) LLaMA, 7 - 65B (Touvron et al., 2023); b) LLaMA 3, 8 and 70B (Grattafiori et al., 2024); c) GLM, 130B (Zeng et al., 2023); d) Bloomz and Bloom 7B (Muenighoff et al., 2023); e) cohere²; f) GPT 3.5 series, including davinci 001-003 model³ and gpt-3.5-turbo (Schulman et al., 2022); g) GPT-4 (OpenAI, 2024a) and GPT-4-o (OpenAI, 2024b); h) a line of models, based on T5 (Xue et al., 2021) and T0 (Sanh et al., 2022); i) Gemma, 7B (Team, 2024b) and Gemma 2, 9B (Team, 2024a); j) GPT-J, 6B (Wang and Komatsuzaki, 2021) and GPT-Neo-X, 20B (Black et al., 2022); k) Mixtral, 8 x 7B (Jiang et al., 2024); l) OPT, 125M - 30B (Zhang et al., 2022).

After analyzing the dataset manually, we identified that some samples contain anomalous punctuation. Figures 5 and 7 display several fragments of such samples. For comparison, Figures 6 and 8 show samples from the same models (or human texts) without these anomalies. We hypothesize that this inconsistency arises from the COLING dataset being composed of multiple datasets created by different authors.

Previous research works have shown that spurious features related to the text length (Kushnareva et al., 2024) and formatting (Dugan et al., 2024) significantly affect artificial text detection. Moreover, Cai and Cui (2023) found that sometimes adding even a single space before the comma may confuse detectors. Thus, we find it important to analyze the peculiarities of the dataset we use and investigate whether the features we examine truly reflect inherent properties of the generated texts or are simply influenced by superficial traits.

Figures 9 and 10 illustrate the frequency of various anomalies across the model generations. In particular, we found that GPT-NeoX generations contain the "...." anomaly most frequently among all models. Meanwhile, human-generated texts in the COLING dataset commonly contain spaces before commas or commas after line breaks, which is likely a side effects of preprocessing procedures applied when the datasets were compiled. Additionally, we discovered that the GPT-4-o model used double line breaks in almost every text it generated; models from the Gemma and LLaMA-3

²<https://docs.cohere.com/docs/models>

³<https://platform.openai.com/docs/models>

families displayed double line breaks in more than half of their generations as well. In contrast, human texts contained far fewer double line breaks, with occurrences of three or more line breaks being relatively rare across all models.

Talking about the lengths of the samples, we see that they also vary a lot (see Figure 11). In particular, T5- and T0- based models tend to generate much shorter texts than other models. Due to this, we investigate further which features are the most sensitive to the length of the input texts and syntactic anomaly in the Appendix D.

C RAID dataset: additional details

RAID dataset contains generations of numerous models, such as GPT-2-XL (Radford et al., 2019), davinci-002⁴, ChatGPT (Schulman et al., 2022), GPT-4 (OpenAI, 2024a), Cohere⁵, Mistral 7B (Jiang et al., 2023), MPT-30B⁶ and LLaMA (Touvron et al., 2023). However, for our purposes we used only the most powerful ones: ChatGPT and GPT-4.

Authors experimented with two types of decoding (greedy and sampling) and applied repetition penalty to a half of generations. Also they applied various types of attacks to the texts, such as:

- **Alternative spelling** (British)
- **Article** ('the', 'a', 'an') **deletion**
- **Adding paragraph** (\n\n) between sentences
- Swapping the case of words from **upper** to **lower** and vice versa
- **Zero-width space**: Inserting the zero-width space U+200B every other character
- Adding **whitespaces** between characters
- **Homoglyph**: Swapping characters for alternatives that look similar
- Randomly shuffling digits of **numbers**
- Inserting common **misspellings**
- **Paraphrasing** with DIPPER (Krishna et al., 2023)
- Replacing words with **synonyms**.

The dataset contains 2,000 continuations for every combination of domain, model, decoding, penalty, and adversarial attack in total. However, for our purposes, we used only 100 continuations for every combination.

⁴<https://platform.openai.com/docs/models>

⁵<https://docs.cohere.com/docs/models>

⁶<https://www.databricks.com/blog/mpt-30b>

Figures 12 and 13 present examples of GPT-4 generations from RAID dataset with and without an attack for comparison.

D Isolating features most sensitive to the length of samples, syntactic anomalies and attacks

To identify the features that are the most sensitive to particular peculiarities of the texts, we took measures to isolate influence of those peculiarities from other text properties, such as the style or topic. To achieve this, we performed the algorithms described below.

D.1 Length

To identify features most sensitive to sample length, we used human-written texts from the COLING dataset (see Appendix B), because COLING contains a significantly larger proportion of human texts compared to model-generated ones, and these texts are much more diverse. Then, we selected those domains of human texts that contain a sufficiently large amount of text samples (> 1000 samples). For each such domain, we identified the top 10% longest and top 10% shortest texts. For both sets, we calculated the values of each feature, then computed the difference between the average feature values for the longest and shortest texts. Thus, for each domain, we identified the top-10 features with the greatest differences. Subsequently, we computed the intersection of these top-10 features across all domains, to eliminate the influence of properties of each particular domain.

D.2 Syntactic anomalies

For each syntactic anomaly, we identified the top three domains of human texts from COLING that contained the highest proportion of texts exhibiting the given anomaly. For each domain, we calculated average feature values for texts with and without the anomaly. Then, we selected top 10 features with the greatest differences for each domain. Finally, we computed the intersection of these top 10 features across all top-3 domains, isolating those features that consistently exhibited the highest sensitivity to the given anomaly. The process was repeated for several layers of SAE.

The results are presented in the Table 1.

As one can see, the most anomalies persistently activate from 1 to 3 SAE features on each layer. However, this method didn't reveal any features